

The artifact Wrapify ML (or Rapify) is a completely new project that is designed to display the listening history of a Spotify user. It was conceptualized a couple weeks back but work on it only began on Monday, November 17th, 2025.

I chose this artifact for my ePortfolio because I felt it shows my versatile skillset and is relevant to some of the modern needs of different tech industries. This artifact displays my ability to work with large datasets and shows my thought process during the cleaning and validation cycle.

One of the challenges I faced was regarding the use of Jupyter Notebook. I wanted my prompts to be accessible to others which would require a physical code to run. While Jupyter lab/notebook seemed an obvious solution at first, it became clear that my large dataset was a bit too intense and it kept freezing. I instead downloaded R and Jupyter notebook packages to visual studio code and was able to work without issue. After that, things were quite simple until I separated my listening history into different years and started to work with the data.

R has great community support, and I was able to find resources very quickly that helped me achieve my desired goals. For example, when I got stuck on how to go about getting the totals for each listening year, I found this resource on [counting using the dplyr package](#) which made the process incredibly simple. I then found an example of the [kable package](#) that explained how to improve the look of my output, making it legible to end users.

I'm dissatisfied with my project, however, because I wanted to achieve these results through machine learning, but that proved incredibly difficult since I am not using numerical data. I was, however, able to use machine learning to predict skipping patterns. Using the OneR package, I was able to determine the likelihood of a song or artist being skipped with over 75% accuracy for

both parameters. While it would have been nice to get around 90% accuracy, that I know can be achieved by using a smaller dataset.

Nevertheless, I met the course outcomes “*Design and evaluate computing solutions that solve a given problem using algorithmic principles and computer science practices and standards appropriate to its solution while managing the trade-offs involved in design choices.*” by my use of machine learning packages like OneR, and “*Design, develop, and deliver professional-quality oral, written, and visual communications that are coherent, technically sound, and appropriately adapted to specific audiences and contexts.*” By my interpretation of the results in a user friendly way, such as through tables.

References:

Machine learning, explained | MIT Sloan. (2021, April 21). MIT Sloan.

<https://mitsloan.mit.edu/ideas-made-to-matter/machine-learning-explained>